

André João César Costa

PhD Student @ Carnegie Mellon University & NOVA School of Science and Technology

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Education

PhD in Computer Science (Dual Degree) — 2024 – Present
CMU Portugal Dual Degree PhD Program

Carnegie Mellon University, Pittsburgh, PA, USA
Advised by Prof. Guy Blelloch

NOVA School of Science and Technology, Lisbon, Portugal
Advised by Prof. João Lourenço

MSc in Computer Science — 2021 – 2023
NOVA School of Science and Technology, Lisbon, Portugal
Advised by Prof. João Lourenço and Prof. Nuno Preguiça
Thesis: *FLeeC: A Fast and Lock-Free Applicational Cache*

BSc in Computer Science — 2018 – 2021
Faculty of Sciences of the University of Lisbon, Lisbon, Portugal

Awards & Funding

Best Student Award (Distinção de Melhor Aluno) — Nov 2024
MSc in Computer Science (MEI), academic year 2022/2023, NOVA School of Science and Technology

PhD Funding, CMU Portugal — 2024 – Present
Dual Degree PhD Program, Carnegie Mellon University and NOVA School of Science and Technology

Publications

Conference Papers

Zachary Kent, Guy E. Blelloch, **André Costa**. **uSTM: A Lightweight and Efficient STM Supporting General Types and Deferred Aborts**. *SPAA 2026 (38th ACM Symposium on Parallelism in Algorithms and Architectures), pp. 102–115. doi:10.1145/3816782.3819198*.

Under Review

André Costa, Guy E. Blelloch, João Lourenço, Zachary Kent. **deqalloc: the memory-stealing allocator**. *Under review*.

National Conferences

Samuel C. Fernão Pires, **André Costa**, João M. Lourenço. **FLeeC+: A Fully Lock-Free High-Performance Caching System**. *INForum 2026 (17th Simpósio de Informática)*.

André J. Costa, Nuno M. Preguiça, João M. Lourenço. **FLeeC: a Fast Lock-Free Application Cache**. *INForum 2023 (14th Simpósio de Informática), pp. 153–155. arXiv:2406.09428*.

Research Projects

Memory Allocation for Concurrent Data Structures
deqalloc: memory-stealing instead of remote frees, for fast allocation under contention (under review).

Software Transactional Memory that is Fast and Usable
uSTM: a lightweight STM supporting general types and deferred aborts (SPAA 2026).

Safe Memory Reclamation while Bounding Memory Consumption
harp: hazard pointers over memory regions (Hazard Region Pointers), for bounded memory at EBR speed (in progress).

Lock-free Key-Value Stores
FLeeC / FLeeC+: a lock-free drop-in replacement for Memcached from my MSc thesis, extended in FLeeC+ (INForum 2023, 2026).

Service

Shadow PC Member, SPAA 2027 — 2027
ACM Symposium on Parallelism in Algorithms and Architectures

Supervision

Samuel C. Fernão Pires, MSc thesis — May 2026
Extending FLeeC: A Non-Blocking Application Cache, NOVA School of Science and Technology
Co-advised with Prof. João Lourenço. Grade 19/20; published at INForum 2026.

Teaching

All as Invited Teaching Assistant at NOVA School of Science and Technology.

	Cloud Computing (MSc)	Sep 2026 – Jan 2027
	Computer Architecture (BSc)	Mar 2026 – Jul 2026
	Concurrent Systems (MSc)	Sep 2025 – Jan 2026
	Introduction to Programming for Science and Engineering (BSc)	Feb 2024 – Jun 2024
	Informatics for Science and Engineering (BSc)	Mar 2023 – Jul 2023
	Introduction to Programming to Science and Engineering (BSc)	Sep 2022 – Jan 2023

Professional Experience

Embedded-Software Engineer — Oct 2023 – Feb 2024
Bosch Car Multimedia Portugal, Braga, Portugal

Skills

Languages: C++, C, Java, Python, Bash
Tools: perf, Valgrind, GDB, AddressSanitizer, CMake, Docker, Git, LaTeX

Languages

Portuguese (native), English (fluent)